

Fixed-Discriminant Gauss Reduction: Cycles, Stabilizers, Reversal, and a Finite Determinant

HCS-C364 source-local theorem package

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Abstract

At one positive nonsquare binary-form discriminant we freeze an exact primitive reduced phase space and prove that the Gauss shift is its permutation. Every orbit has a primitive continued-fraction word whose positive matrix product is a hyperbolic stabilizer; the same matrix gives an exact cycle multiplier and its discriminant identity. Finite exact computations audit conventions only; the theorem is analytic.

1 The convention and the finite phase space

Fix a positive nonsquare integer $\Delta \equiv 0, 1 \pmod{4}$. Let $\mathcal{R}_\Delta$ contain the numbers

$$\alpha = \frac{P + \sqrt{\Delta}}{Q}, \quad Q = 2A > 0, \quad A = \frac{Q}{2}, \quad B = -P, \quad C = \frac{P^2 - \Delta}{2Q}, \quad (1)$$

for which $A, B, C \in \mathbb{Z}$, $\gcd(A, B, C) = 1$, and

$$\alpha > 1, \quad -1 < \bar{\alpha} = \frac{P - \sqrt{\Delta}}{Q} < 0. \quad (2)$$

The polynomial $Ax^2 + Bx + C$ has discriminant exactly Δ and roots $(P \pm \sqrt{\Delta})/Q$. Thus (1) is a binary-form discriminant convention, not a radicand convention that accidentally changes the discriminant to 4Δ .

Define

$$G(\alpha) = \frac{1}{\alpha - \lfloor \alpha \rfloor}. \quad (3)$$

Theorem 1 (Complete reduced dynamics). *The set $\mathcal{R}_\Delta$ is finite and G is a bijection. Let $\alpha_0, \dots, \alpha_{\ell-1}$ be a minimal cycle and $a_j = \lfloor \alpha_j \rfloor$. Let $A(a)$ denote the two-by-two matrix with rows $(a, 1)$ and $(1, 0)$. Then*

$$M = A(a_0)A(a_1) \cdots A(a_{\ell-1}). \quad (4)$$

Writing $M_{11} = a, M_{12} = b, M_{21} = c, M_{22} = d$, this matrix fixes α_0 , has $\det M = (-1)^\ell$, and is the primitive positive inverse-branch stabilizer. For some integer $k > 0$,

$$(c, d - a, -b) = k(A, B, C), \quad (\operatorname{tr} M)^2 - 4 \det M = k^2 \Delta. \quad (5)$$

If $\lambda = c\alpha_0 + d$, then $\lambda > 1$ and

$$|(G^\ell)'(\alpha_0)| = \lambda^2. \quad (6)$$

Proof. From (2), $|P| < \sqrt{\Delta}$ and $0 < Q < P + \sqrt{\Delta} < 2\sqrt{\Delta}$, proving finiteness. Put $a = \lfloor \alpha \rfloor$,

$$P_1 = aQ - P, \quad Q_1 = \frac{\Delta - P_1^2}{Q}. \quad (7)$$

Rationalization gives $G(\alpha) = (P_1 + \sqrt{\Delta})/Q_1$. Because $0 < \alpha - a < 1$, this root exceeds one; because $-a - 1 < \bar{\alpha} - a < -a$, its conjugate lies in $(-1, 0)$. The original integrality says $2Q \mid P^2 - \Delta$, while $P_1^2 - P^2 = a^2Q^2 - 2aPQ$ is divisible by $2Q$; hence

Q_1 is positive and even and the new form is integral. The inverse branch $x \mapsto a + 1/x$ is unimodular, so form content, and hence primitivity, is preserved.

Conversely, for reduced β , set

$$t = -1/\bar{\beta}, \quad a = \lfloor t \rfloor, \quad \alpha = a + 1/\beta. \quad (8)$$

Here $t > 1$ is irrational, so $\alpha > 1$ and $\bar{\alpha} = a - t \in (-1, 0)$. The inverse unimodular substitution preserves the integral primitive form and $G(\alpha) = \beta$. The conjugate interval determines a uniquely, proving bijectivity.

Writing $H_a(x) = a + 1/x$, one has $\alpha_j = H_{a_j}(\alpha_{j+1})$; therefore (4) fixes α_0 . A shorter digit word would, by uniqueness of an infinite simple continued fraction, give a shorter state return. Each factor has determinant minus one. The integral stabilizer of a real quadratic irrational maps through $N \mapsto c_N \alpha + d_N$, after choosing the positive representative, to units of its quadratic order that are positive at the α -embedding. We use the classical reduction–automorph cross-section lemma in its exact return form: an expanding positive projective stabilizer preserves the oriented geodesic from $\bar{\alpha}$ to α and advances its reduced cross-section by an integer $r \geq 1$ of complete Gauss digits. Thus

$$[N] = [A(a_0) \cdots A(a_{r-1})], \quad G^r(\alpha_0) = \alpha_0,$$

and every complete return gives such a stabilizer. This standard continued-fraction correspondence is source-anchored to Uspensky and Buell below. Minimality of ℓ makes every $r = n\ell$, and periodicity makes the corresponding product M^n ; a representative first written with mixed signs therefore cannot yield a smaller positive return. Hence M generates the positive projective stabilizer. Since $\det(M^n) = (-1)^{\ell n}$, the determinant-one stabilizer up to $\pm I$ is generated by M for even ℓ and by M^2 for odd ℓ .

The fixed-point equation of M is $cx^2 + (d - a)x - b = 0$. Its coefficient vector is an integral multiple of the primitive vector (A, B, C) , giving (5). Finally $M'(\alpha) = \det M / (c\alpha + d)^2$; since M is the local inverse of G^ℓ , reciprocal differentiation gives (6). Positive digit matrices give $\lambda > 1$. This is the *reduced-permutation owner*. QED

Source and collision boundary

Uspensky’s reduction paper appears in the *Bulletin of the AMS* (1930), 710–718, doi:10.1090/S0002-9904-1930-05043-0. Buell’s chapter “Indefinite Forms” fixes authoritative form conventions, doi:10.1007/978-1-4612-4542-1_3. Zagier supplies real-quadratic arithmetic context, doi:10.1007/978-3-642-61829-1. These sources establish lineage, not priority for this packaging. C16 uses continued fractions in an S-arithmetic clock, C193 is a Markoff–Vieta descent tree, and C330 is Romik’s Pythagorean map; none owns this fixed-discriminant finite reduction permutation.